

Пример 01.04.2026

16. задача

1.1

$$Z = 2x_1 + 4x_2 - 2x_3 \rightarrow \min$$

$$\begin{cases} x_1 + 2x_2 - x_3 \geq 1 \\ 2x_1 - x_2 + 3x_3 \leq 3 \quad | \cdot (-1) \\ x_1, x_2, x_3 \geq 0 \end{cases}$$

$$\begin{cases} x_1 + 2x_2 - x_3 \geq 1 \\ -2x_1 + x_2 - 3x_3 \geq -3 \\ x_1, x_2, x_3 \geq 0 \end{cases}$$

$$y_1 \begin{pmatrix} x_1 & x_2 & x_3 \\ 1 & 2 & -1 \\ -2 & 1 & -3 \end{pmatrix}$$

$$x_1 \begin{pmatrix} y_1 & y_2 \\ 1 & -2 \\ 2 & 1 \\ -1 & -2 \end{pmatrix}$$

$$\begin{cases} y_1 - 2y_2 \leq 2 \\ 2y_1 + y_2 \leq 10 \\ -y_1 - 2y_2 \leq -2 \\ y_1, y_2 \geq 0 \end{cases}$$

$$W = y_1 - 3y_2 \rightarrow \max$$

1.2

$$Z = 2x_1 + 3x_2 - x_3 + 5x_4 \rightarrow \max$$

$$\begin{cases} 5x_1 + 4x_2 + 3x_3 - x_4 = 7 \\ x_1 + 2x_2 - x_3 \leq 5 \\ 2x_1 + x_2 + x_3 + 3x_4 = 8 \end{cases}$$

$$y_1 \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \\ 5 & 4 & 3 & -1 \\ 1 & 2 & -1 & 0 \\ 2 & 1 & 1 & 3 \end{pmatrix}$$

=>

$$x_1 \begin{pmatrix} y_1 & y_2 & y_3 \\ 5 & 1 & 2 \\ 4 & 2 & 1 \\ 3 & -1 & 1 \\ -1 & 0 & 3 \end{pmatrix}$$

$$\begin{cases} 5y_1 + y_2 + 2y_3 = 2 \\ 2y_1 + 2y_2 + y_3 = 5 \\ 3y_1 - y_2 + y_3 = -1 \\ -y_1 + 5y_3 = 5 \\ y_2 \geq 0 \end{cases}$$

$$W = 2y_1 + 5y_2 + 8y_3 \rightarrow \min$$

$\sqrt{3}$

$$Z = 7x_1 + 6x_2 + 3x_3 - x_4 \rightarrow \min$$

$$\begin{cases} 2x_1 - x_2 + 2x_3 - 3x_4 \geq 12 \\ -x_1 + 2x_2 - x_3 + x_4 \leq 10 \quad | \cdot -1 \quad x_1 - 2x_2 + x_3 - x_4 \geq -10 \\ 3x_1 + 5x_2 + 4x_4 = 7 \\ x_2 \geq 0, x_3 \geq 0 \end{cases}$$

$$\begin{array}{c|cccc} & x_1 & x_2 & x_3 & x_4 \\ \hline y_1 & 2 & -1 & 2 & -3 \\ y_2 & -1 & 2 & -1 & 1 \\ y_3 & 3 & 5 & 0 & 4 \end{array}$$

$$\Rightarrow \begin{array}{c|ccc} & y_1 & y_2 & y_3 \\ \hline x_1 & 2 & -1 & 2 \\ x_2 & -1 & 2 & -1 \\ x_3 & 2 & -1 & 2 \\ x_4 & -3 & 1 & 4 \end{array}$$

$$\begin{cases} 2y_1 + y_2 + 3y_3 = 7 \\ -y_1 - 2y_2 - 5y_3 \leq 6 \\ 2y_1 + y_2 \leq 3 \\ -3y_1 - y_2 + 4y_3 \leq -1 \\ y_1, y_2 \geq 0 \end{cases}$$

$$W = 12y_1 - 10y_2 + 7y_3 \rightarrow \min$$

$$\begin{array}{c|ccc} & y_1 & y_2 & y_3 \\ \hline x_1 & 2 & -1 & 2 \\ x_2 & -1 & 2 & -1 \\ x_3 & 2 & -1 & 2 \\ x_4 & -3 & 1 & 4 \end{array}$$

1/7

$$Z = 6x_1 + 4x_2 + 8x_3 + 2x_4 - 3x_5 \rightarrow \max$$

$$\begin{cases} 2x_1 + 2x_2 + 2x_3 + 2x_4 + x_5 \leq 7 \\ x_1 - 2x_3 + x_4 + 2x_5 \geq 8 \end{cases} \quad | \cdot (-1) \quad | -x_1 + 2x_3 - x_4 - 2x_5 = -8$$

$$x_i \geq 0, i = 1, \dots, 5$$

$$\begin{pmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \\ y_1 & 2 & 2 & 2 & 1 \\ y_2 & -1 & 0 & -1 & -2 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 & y_1 & y_2 \\ x_2 & 2 & 0 \\ x_3 & 2 & -2 \\ x_4 & 2 & -1 \\ x_5 & 1 & -2 \end{pmatrix}$$

$$\begin{cases} 2y_1 - y_2 \geq 6 \\ 2y_1 \geq 4 \\ 2y_1 + 2y_2 \geq 8 \\ 2y_1 - y_2 \geq 2 \\ y_1 - 2y_2 \geq -3 \\ y_1, y_2 \geq 0 \end{cases}$$

$$W = 7y_1 - 8y_2 \rightarrow \min$$

1) $2y_1 - y_2 = 6$

y_1	0	3
y_2	-6	0

2) $2y_1 = 4$

y_1	0	2
y_2	4	0

3) $2y_1 + 2y_2 = 8$

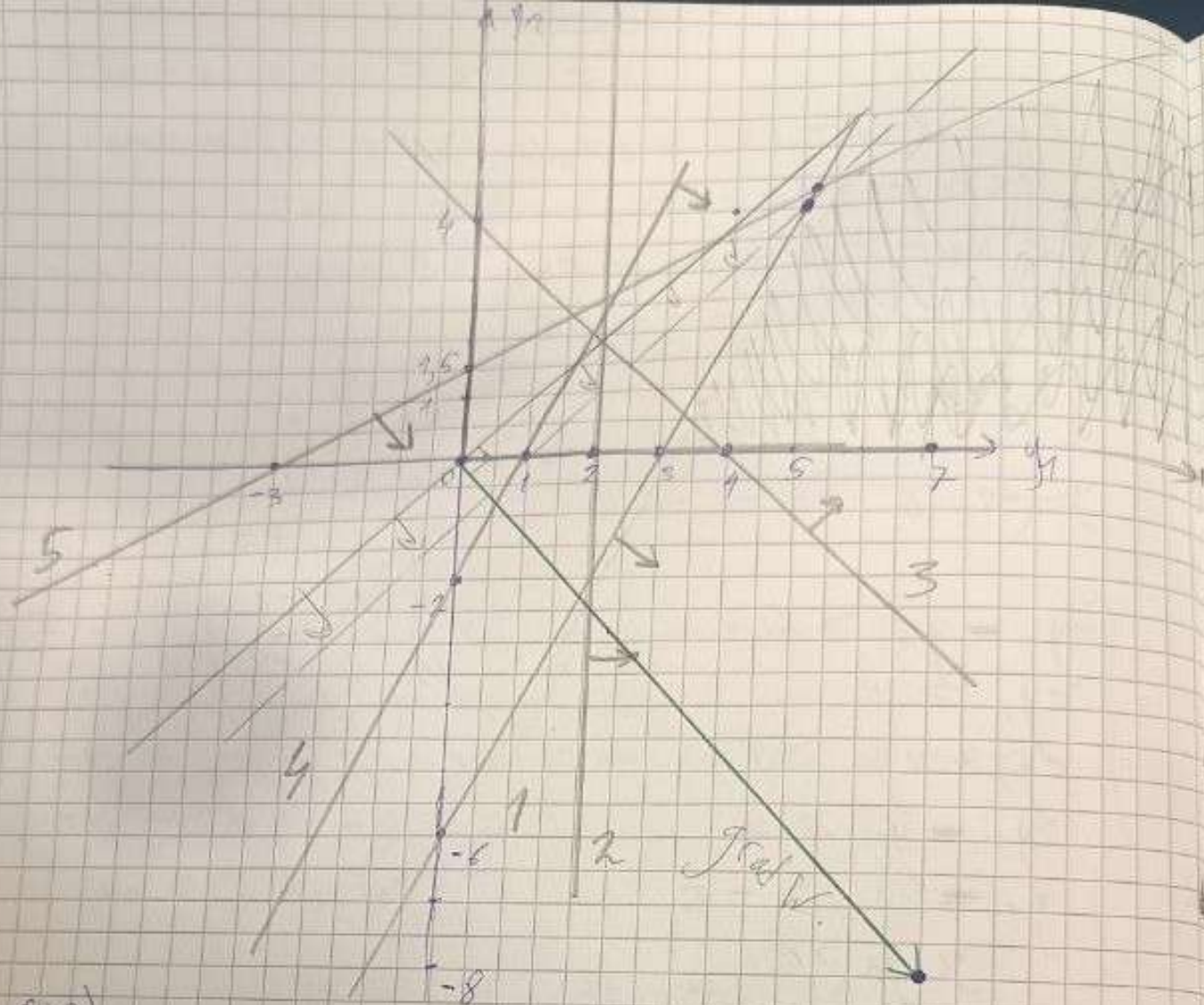
y_1	0	4
y_2	4	0

4) $2y_1 - y_2 = 2$

y_1	0	1
y_2	-2	0

5) $y_1 - 2y_2 = -3$

y_1	0	-3
y_2	1.5	0



$(0,0)$

1) $0 \geq 6 -$ 4) $0 \geq 2 -$

2) $0 \geq 4 -$ 5) $0 \geq -3 +$

3) $0 \geq 8 -$

$(5,4)$

1) $2 \cdot 5 - 4 = 6 \geq 6 +$

2) $2 \cdot 5 \geq 10 \geq 4 +$

3) $2 \cdot 5 + 2 \cdot 4 = 18 \geq 8 +$

4) $2 \cdot 5 - 4 = 6 \geq 2 +$

5) $5 - 2 \cdot 4 = 5 - 8 = -3 \geq -3 +$

$\text{Max } W_{\text{min}} = 7 \cdot 5 - 8 \cdot 4 = 35 - 32 = 3$

$$\begin{cases} y_1 \cdot (7 - 2x_1 - 2x_2 - 2x_3 - 2x_4 - x_5) = 0 \\ y_2 \cdot (-8 + x_1 - 2x_2 + x_4 + 2x_5) = 0 \\ x_1 \cdot (1 \cdot y_1 - y_2 - 6) = 0 \\ x_2 \cdot (2 \cdot y_1 - 4) = 0 \\ x_3 \cdot (2 \cdot y_1 + 2 \cdot y_2 - 8) = 0 \\ x_4 \cdot (2 \cdot y_1 - y_2 - 2) = 0 \\ x_5 \cdot (y_1 - 2 \cdot y_2 + 3) = 0 \end{cases}$$

$$x_5 \cdot (5 - 2 \cdot 4 + 3) = 0 \quad x_5 \cdot (5 - 8 + 3) =$$

$$= x_5 \cdot 0$$

$$x_5 \neq 0$$

$$x_4 \cdot (2 \cdot 5 - 4 - 2) = x_4 \cdot (10 - 6) = x_4 \cdot 4 = 0$$

$$x_4 = 0$$

$$x_3 \cdot (2 \cdot 5 + 2 \cdot 4 - 8) = x_3 \cdot (10 + 8 - 8) =$$

$$= x_3 \cdot 10 = 0$$

$$x_3 = 0$$

$$x_2 \cdot (2 \cdot 5 - 4) = 0$$

$$x_2 \cdot (10 - 4) = x_2 \cdot 6 = 0$$

$$x_2 = 0$$

$$x_1 \cdot (2 \cdot 5 - 4 - 6) = x_1 \cdot (10 - 10) = x_1 \cdot 0$$

$$x_1 \neq 0$$

$$4 \cdot (1 - 8 + x_1 - 2 \cdot 0 + 0 + 2 \cdot x_5) = 0$$

$$4 \cdot (1 - 8 + x_1 + 2x_5) = 0$$

$$-8 + x_1 + 2x_5 = 0$$

$$5 \cdot (7 - 2x_1 - 2 \cdot 0 - 2 \cdot 0 - 2 \cdot 0 - x_5) = 0$$

$$7 - 2x_1 - x_5 = 0$$

$$x_1 = 8 - 2x_5$$

$$y_1 = 8 - 2 \cdot 3 = 2$$

$$7 - 2(8 - 2x_5) - x_5 = 7 - 16 + 4x_5 - x_5 = 0$$

$$7 - 16 + 3x_5 = 0 \quad 3x_5 = 9 \quad x_5 = 3$$

$$Z_{max} = Z(2; 0; 0; 0; 3) =$$

$$= 1 \cdot 2 + 5 \cdot 0 + 4 \cdot 0 + 2 \cdot 0 - 3 \cdot 3 =$$

$$= 12 - 9 = 3$$

Answer: $Z_{max} = Z(2; 0; 0; 0; 3) = 3$

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$$Z = 3x_1 + 5x_2 + 4x_3 - 5x_4 - 3x_5 \rightarrow \min$$

$$\begin{cases} 2x_1 - x_3 - 2x_4 - x_5 \geq 2 \\ -x_1 + x_2 - 2x_3 + x_4 + x_5 \leq -2 \quad | \cdot -1 \\ x_i \geq 0, i = 1, \dots, 5 \end{cases}$$

$$\begin{cases} 2x_2 - x_3 - 2x_4 - x_5 \geq 2 \\ x_1 - x_2 + 2x_3 - x_4 - x_5 \geq 2 \\ x_i \geq 0, i = 1, \dots, 5 \end{cases}$$

$$\begin{array}{c|ccccc} & x_1 & x_2 & x_3 & x_4 & x_5 \\ y_1 & 0 & 2 & -1 & -2 & -1 \\ y_2 & 1 & -1 & 2 & -1 & -1 \end{array} \Rightarrow \begin{array}{c|cc} & y_1 & y_2 \\ x_1 & 0 & 1 \\ x_2 & 2 & -1 \\ x_3 & -1 & 2 \\ x_4 & -2 & -1 \\ x_5 & -1 & -1 \end{array}$$

$$W = 2y_1 + 2y_2 \rightarrow \max$$

$$y_2 \leq 3$$

$$2y_1 - y_2 \leq 5$$

$$-y_1 + 2y_2 \leq 4$$

$$-2y_1 - y_2 \leq -5$$

$$-y_1 - y_2 \leq -3$$

$$y_1, y_2 \geq 0$$

1) $y_2 = 3$

y_1	0	3
y_2	3	0

2) $2y_1 - y_2 = 5$

y_1	0	2,5
y_2	-5	0

3) $-y_1 + 2y_2 = 4$

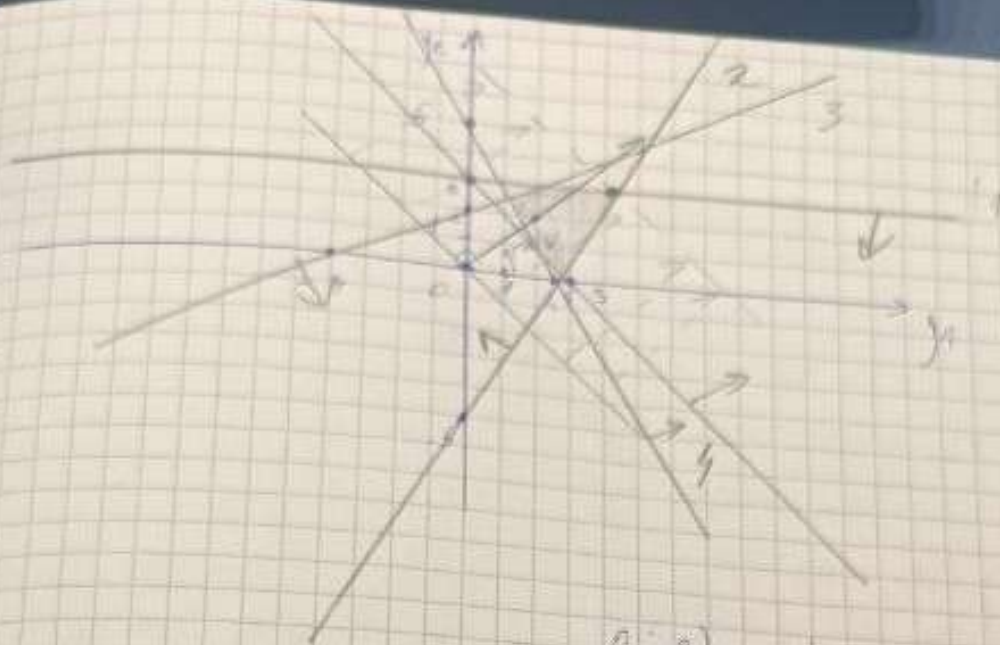
y_1	0	-4
y_2	2	0

4) $-2y_1 - y_2 = -5$

y_1	0	2,5
y_2	5	0

5) $-y_1 - y_2 = -3$

y_1	0	3
y_2	3	0



Optimal: (4; 3)

$$\begin{aligned} 0, 0) \quad & 1) 0 \leq 3 + \quad 4) 0 \leq 5 \\ & 2) 0 \leq 5 + \quad 5) 0 \leq -3 \\ & 3) 0 \leq 4 + \end{aligned}$$

$$\begin{aligned} 1) & 3 \leq 3 + \\ 2) & 2 \cdot 4 - 3 = 8 - 3 = 5 \leq 5 + \\ 3) & -4 + 2 \cdot 3 = -4 + 6 = 2 \leq 4 + \\ 4) & -2 \cdot 4 - 3 = -8 - 3 = -11 \leq -5 + \\ 5) & -4 - 3 = -7 \leq -3 + \end{aligned}$$

$$W_{\max} = 2 \cdot 4 + 2 \cdot 3 = 8 + 6 = 14$$

$$x_1 \neq 0 \quad x_2 \neq 0$$

$$x_3 = 0 \quad x_4 = 0 \quad x_5 = 0$$

$$\begin{cases} y_1 \cdot (2 - 2x_2) = 0 \\ y_2 \cdot (2 - x_1 + x_2) = 0 \end{cases}$$

$$\begin{cases} 4 \cdot (2 - 2x_2) = 0 \\ 3 \cdot (2 - x_1 + x_2) = 0 \end{cases}$$

$$x_1 = 2 + 1 = 3$$

$$\begin{cases} 2 - 2x_2 = 0 & -2x_2 = -2 \\ 2 - x_1 + x_2 = 0 & x_2 = 1 \end{cases}$$

$$Z_{\min} = Z(3; 1; 0; 0; 0) = 3 \cdot 3 + 5 \cdot 1 + 0 - 0 - 0 = 9 + 5 = 14$$

Answer: $Z_{max} = Z(2; 1; 0; 0; 0) = 14$

Двойственная задача $\sqrt{5}$

$\sqrt{4}$

$$Z = x_1 + x_2 + 2x_3 \rightarrow \max$$

$$\begin{cases} 3x_1 + 4x_2 - x_3 = 4 \\ x_1 - x_2 + 2x_3 = 6 \\ 2x_1 + x_3 = 5 \\ x_i \geq 0, i = 1, 2, 3 \end{cases}$$

$$\begin{matrix} x_1 & x_2 & x_3 \\ y_1 & \begin{pmatrix} 3 & 4 & -1 \end{pmatrix} \\ y_2 & \begin{pmatrix} 1 & -1 & 2 \end{pmatrix} \\ y_3 & \begin{pmatrix} 2 & 0 & 1 \end{pmatrix} \end{matrix} \quad \begin{matrix} y_1 & y_2 & y_3 \\ y_1 & \begin{pmatrix} 3 & 1 & 2 \end{pmatrix} \\ y_2 & \begin{pmatrix} 4 & -1 & 0 \end{pmatrix} \\ y_3 & \begin{pmatrix} -1 & 2 & 1 \end{pmatrix} \end{matrix}$$

$$\begin{cases} 3y_1 + y_2 + 2y_3 \geq 1 \\ 4y_1 - y_2 \geq 1 \\ -y_1 + 2y_2 + y_3 \geq 2 \end{cases}$$

$$W = 4y_1 + 6y_2 + 5y_3 \rightarrow \min$$

$\sqrt{6}$

$$Z = x_1 + 5x_2 + x_3 - 2x_4 \rightarrow \max$$

$$\begin{cases} x_1 - 2x_2 + x_4 = 5 \\ 2x_1 - x_2 + 3x_3 \geq 9 \quad | \cdot (-1) & -2x_1 + x_2 - 3x_3 \leq -9 \\ x_1 + x_2 + x_3 - 2x_4 \leq 1 \\ x_1 \geq 0, x_3 \geq 0 \end{cases}$$

$$W \begin{pmatrix} y_1 & y_2 & y_3 & y_4 \\ 1 & -2 & 0 & 1 \\ 2 & -1 & 3 & 0 \\ 1 & 1 & 1 & -2 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & -2 & 1 \\ -2 & -1 & 1 \\ 0 & -3 & 1 \\ 1 & 0 & -2 \end{pmatrix}$$

$$\begin{cases} y_1 - 2y_2 + y_3 \geq 1 \\ -2y_1 + y_2 + y_3 = 5 \\ -3y_2 + y_3 \geq 1 \\ y_1 - 2y_3 = -2 \\ y_2 \geq 0, y_3 \geq 0 \end{cases}$$

$$W = 5y_1 - 9y_2 + y_3$$

$$\sqrt{9}$$

$$Z = 6x_1 - 2x_2 - 4x_3 - 3x_4 \rightarrow \min$$

$$\begin{cases} x_1 - x_2 - 2x_3 - x_4 \geq -3 \\ -x_1 + x_2 - 2x_3 + 2x_4 \leq -13 \quad | \cdot -1 \\ x_i \geq 0, i = 1, \dots, 4 \end{cases}$$

$$y_1 \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \\ 1 & -1 & -2 & -1 \\ -1 & 1 & -2 & 2 \end{pmatrix} \Rightarrow \begin{matrix} y_1 & y_2 \\ x_1 & 1 & +1 \\ x_2 & -1 & -1 \\ x_3 & -2 & +2 \\ x_4 & -1 & -2 \end{matrix}$$

$$\begin{cases} y_1 + y_2 \leq 6 \\ -y_1 - y_2 \leq -2 \\ -2y_1 + 2y_2 \leq -4 \\ -y_1 - 2y_2 \leq -3 \\ y_1, y_2 \geq 0 \end{cases}$$

$$1) y_1 + y_2 = 6$$

$$\begin{array}{cc|c} y_1 & 0 & 6 \\ y_2 & 6 & 0 \end{array}$$

$$2) -y_1 - y_2 = -2$$

$$\begin{array}{cc|c} y_1 & 0 & 2 \\ y_2 & 2 & 0 \end{array}$$

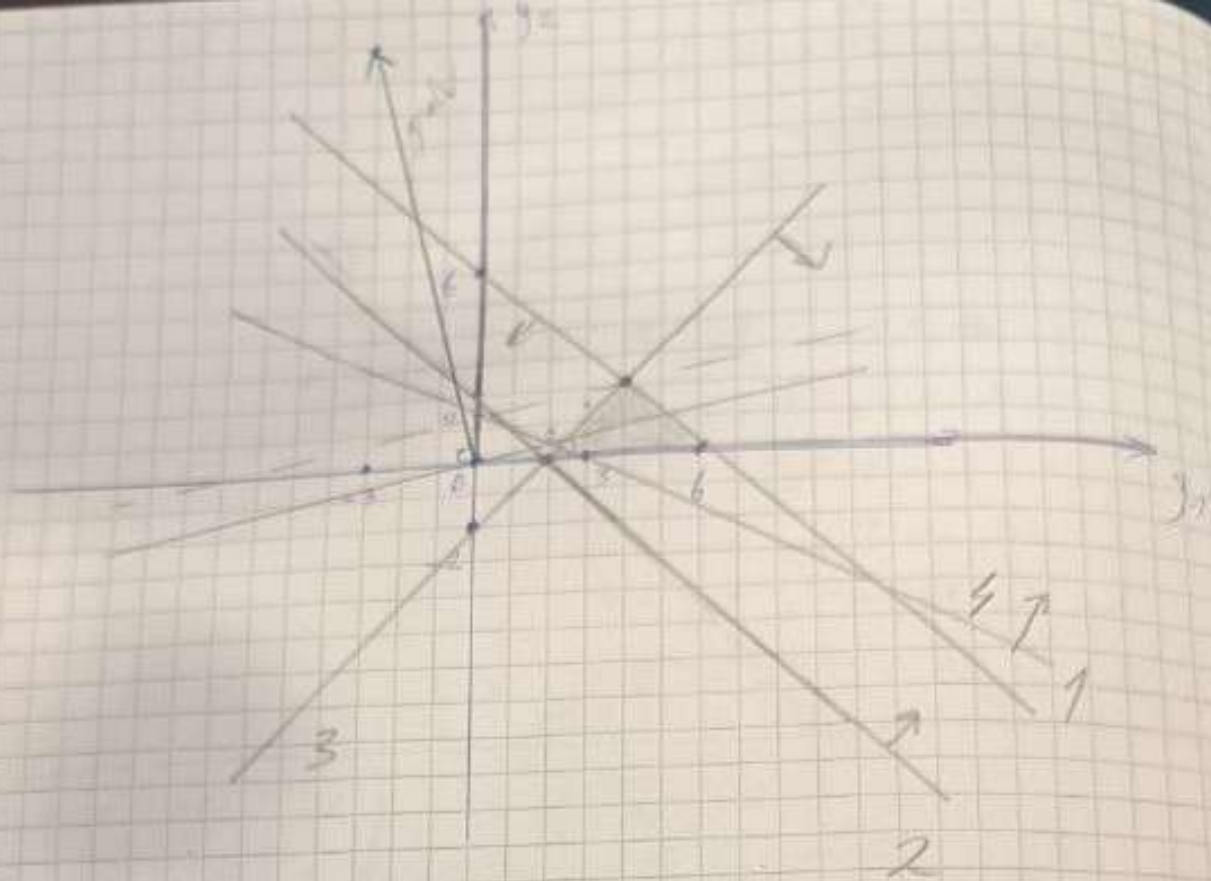
$$3) -2y_1 + 2y_2 = -4$$

$$\begin{array}{cc|c} y_1 & 0 & 2 \\ y_2 & -2 & 0 \end{array}$$

$$W = -3y_1 + 13y_2 \rightarrow \max$$

$$4) -y_1 - 2y_2 = -3$$

$$\begin{array}{cc|c} y_1 & 0 & 3 \\ y_2 & 1.5 & 0 \end{array}$$



$(0,0)$

- 1) $0 \leq 6 +$
- 2) $0 \leq -2 -$
- 3) $0 \leq -4 -$
- 4) $0 \leq -3 -$

$(4,2)$:

- 1) $4 + 2 = 6 \leq 6 +$
- 2) $-4 - 2 = -6 \leq -2 -$
- 3) $-2 \cdot 4 + 2 \cdot 2 = -8 + 4 = -4 \leq -4 -$
- 4) $-4 - 2 \cdot 2 = -4 - 4 = -8 \leq -3 -$

$$1) 4 + 2 = 6 = 6$$

$$3) -2 \cdot 4 + 2 \cdot 2 = -8 + 4 = -4 = -4$$

Terminale: $W(4,2)$

$$W_{\max} = W(4,2) = -3 \cdot 4 + 13 \cdot 2 = -12 + 26 = 14$$

$$x_1 \neq 0, x_3 \neq 0; x_2 = 0, x_4 = 0$$

$$\begin{cases} y_1 \cdot (-3 - x_1 + x_2 + 2x_3 + x_4) = 0 \\ y_2 \cdot (13 - x_1 + x_2 - 2x_3 + 2x_4) = 0 \end{cases}$$

$$\begin{cases} y_1 \cdot (-3 - x_1 + x_2 + 2x_3 + x_4) = 0 \\ y_2 \cdot (13 - x_1 + x_2 - 2x_3 + 2x_4) = 0 \end{cases}$$

$$4 \cdot (-3 - x_1 + 2x_3) = 0$$

$$2 \cdot (13 - x_1 + 2x_3) = 0$$

$$\begin{cases} -3 - X_1 + 2X_3 = 0 \\ 13 - X_1 - 2X_3 = 0 \end{cases}$$

$$X_1 = -3 + 2X_3$$

$$13 - (-3 + 2X_3) - 2X_3 = 13 + 3 - 2X_3 - 2X_3 = 0$$

$$16 - 4X_3 = 0$$

$$X_3 = 4$$

$$X_1 = -3 + 2 \cdot 4 = -3 + 8 = 5$$

$$Z_{\min} = Z(5; 0; 4; 0) = 6 \cdot 5 - 2 \cdot 0 - 4 \cdot 4 = 30 - 16 = 14$$

Answer: $Z_{\min} = Z(5; 0; 4; 0) = 14$